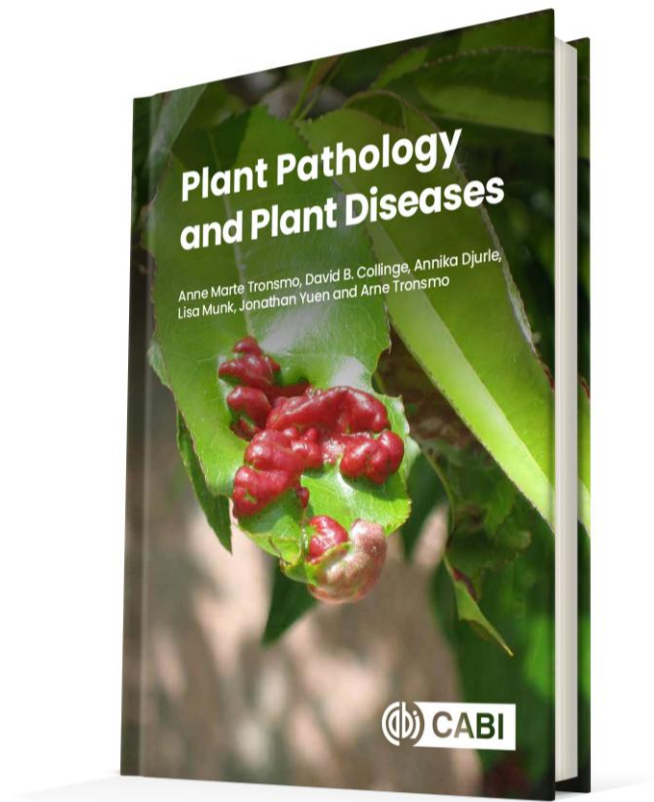




Plant Pathology and Plant Diseases

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Lisa Munk, Jonathan Yuen and Arne Tronsmo 2020

Fig. 20.1 Disease diagnosis and qualitative and quantitative measurements are relevant elements in plant disease assessment. (L. Munk.)

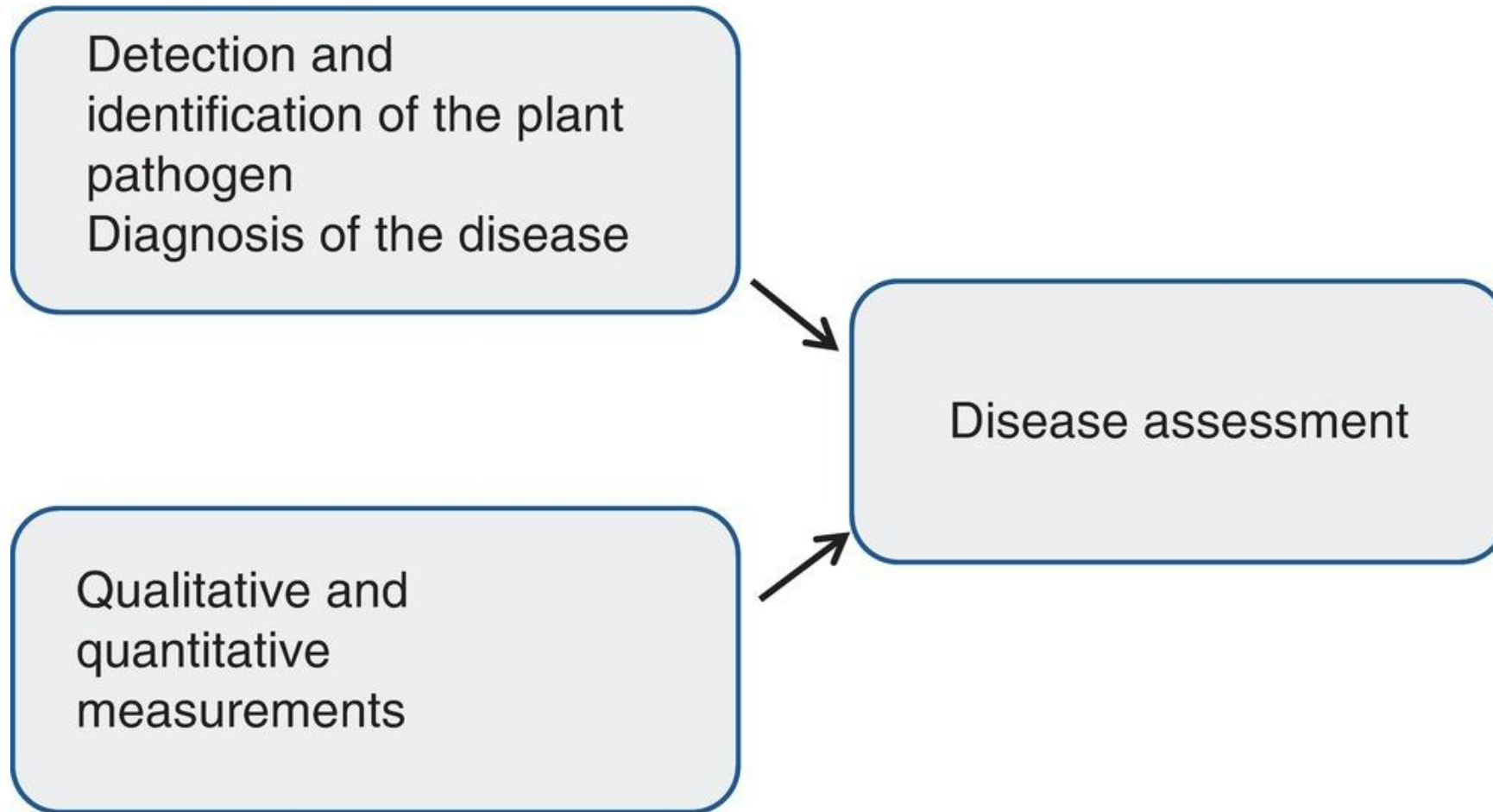


Fig. 20.2 Golfers can be (A) accurate and precise, (B) precise but not accurate and (C) neither accurate nor precise. (© J. Yuen.)

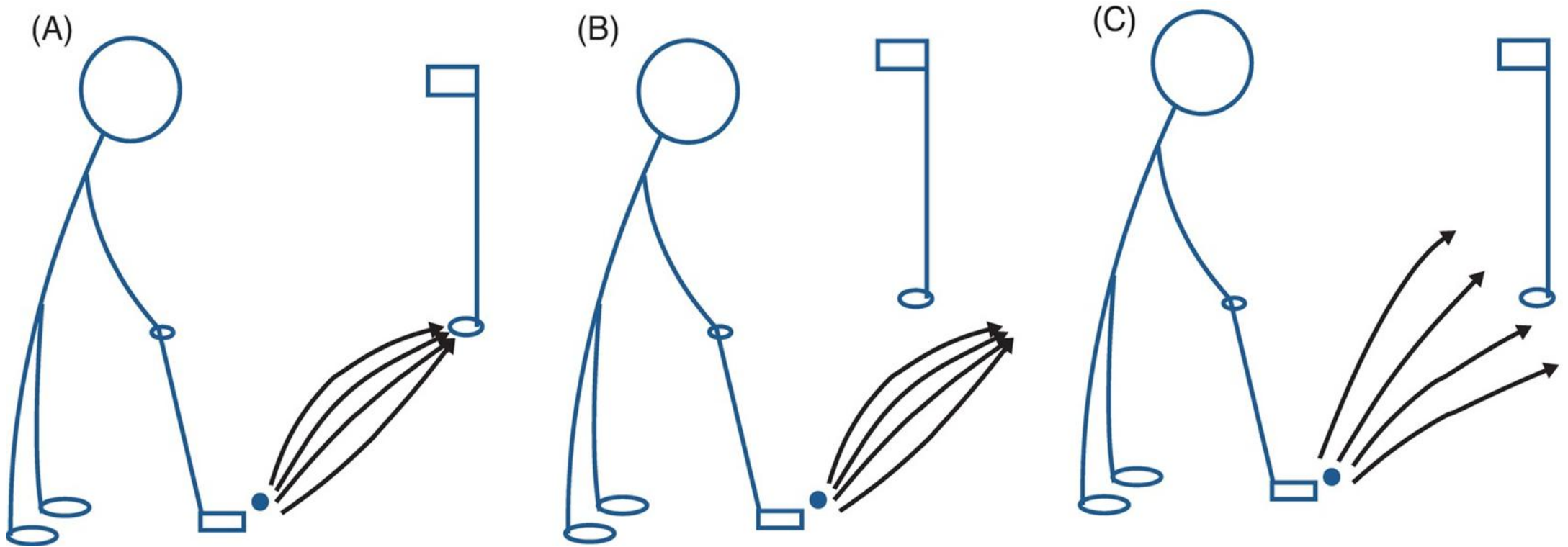


Fig. 20.3 (A) AUDPC is calculated from the added surfaces of rectangles. (B) Two shapes resulting in the same area but not the same epidemic. (A: © Djurle; B: © B. Andersson.)

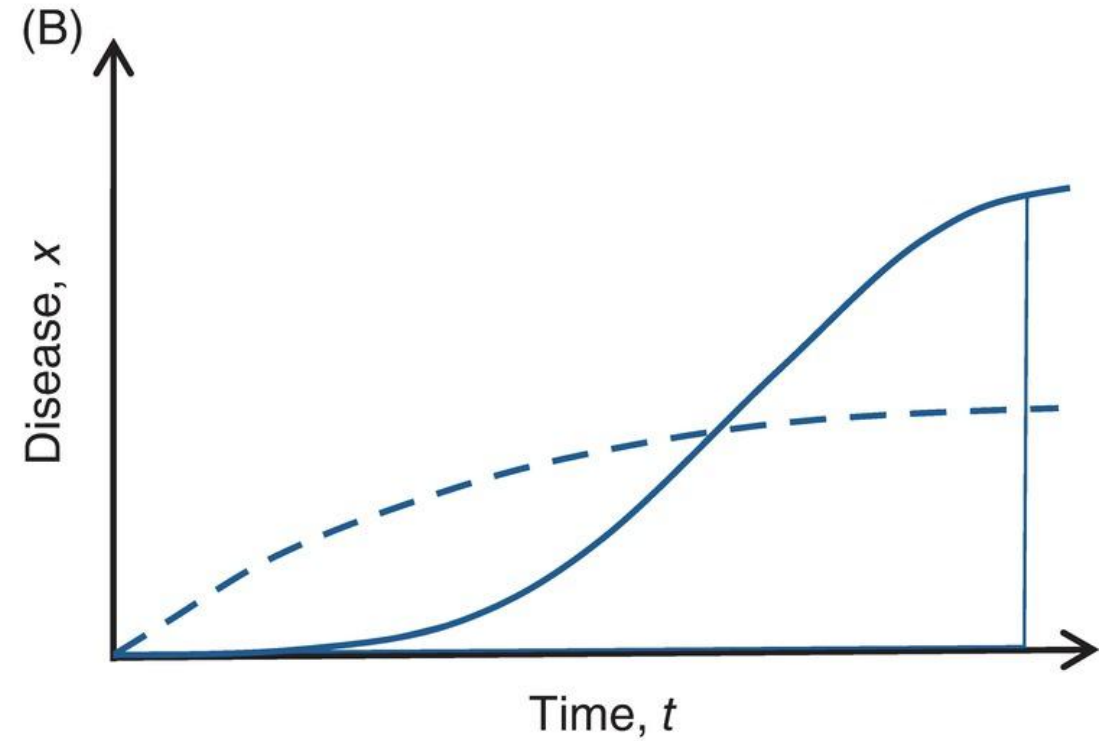
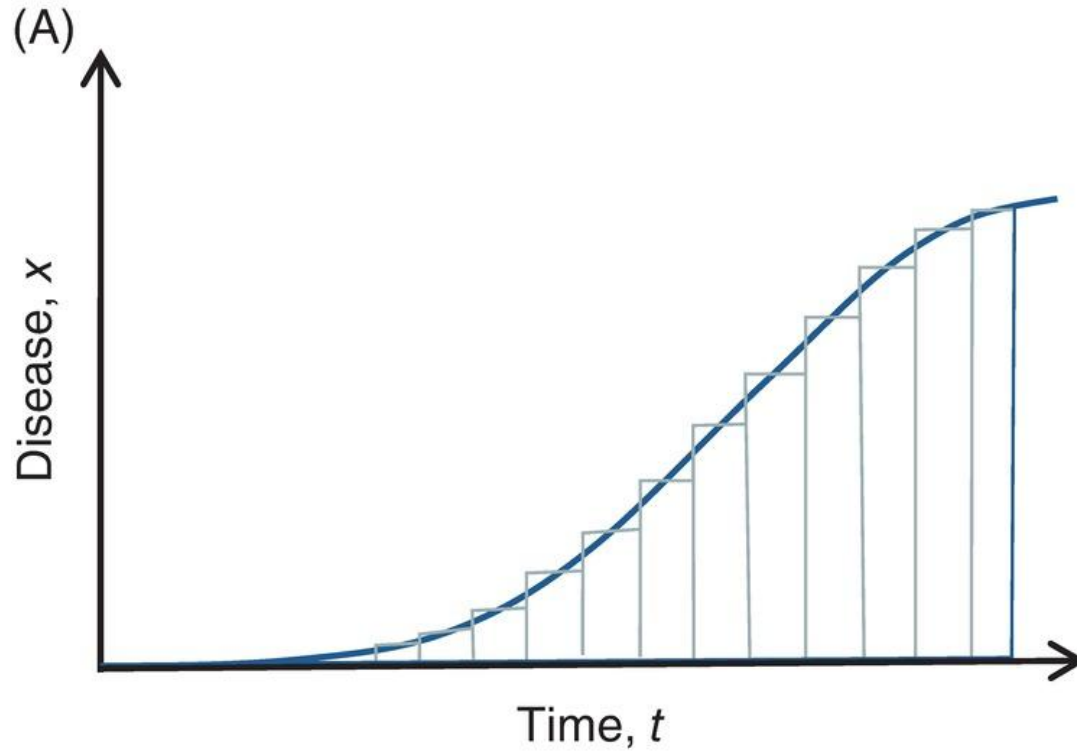


Fig. 20.4 Phenotyping platform.
(© L. Munk.)



Fig. 20.5 Drone (Mavic 2 pro) mounted with a camera. (© L. Munk.)



Fig. 20.6 Standard assessment keys for estimating the severity of leaf spots of cereals and late blight of potato. (James, 1971.)

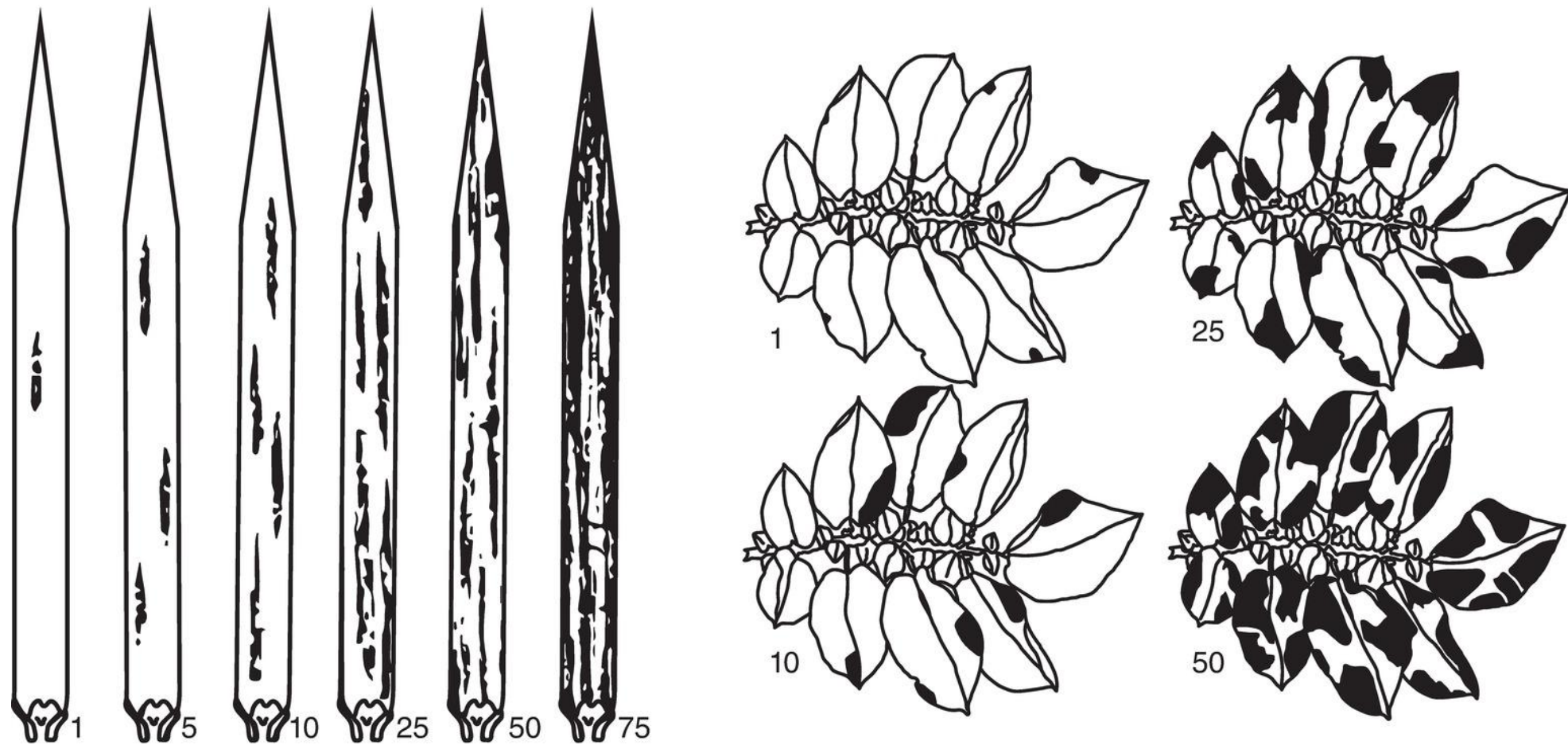


Fig. 20.7 Barley leaves with powdery mildew: (A) type 2 (moderately resistant) and (B) type 4 (susceptible). (© L. Munk.)



Fig. 20.8 Growth stages of pea. (© K. Göransson, courtesy of SLU.)

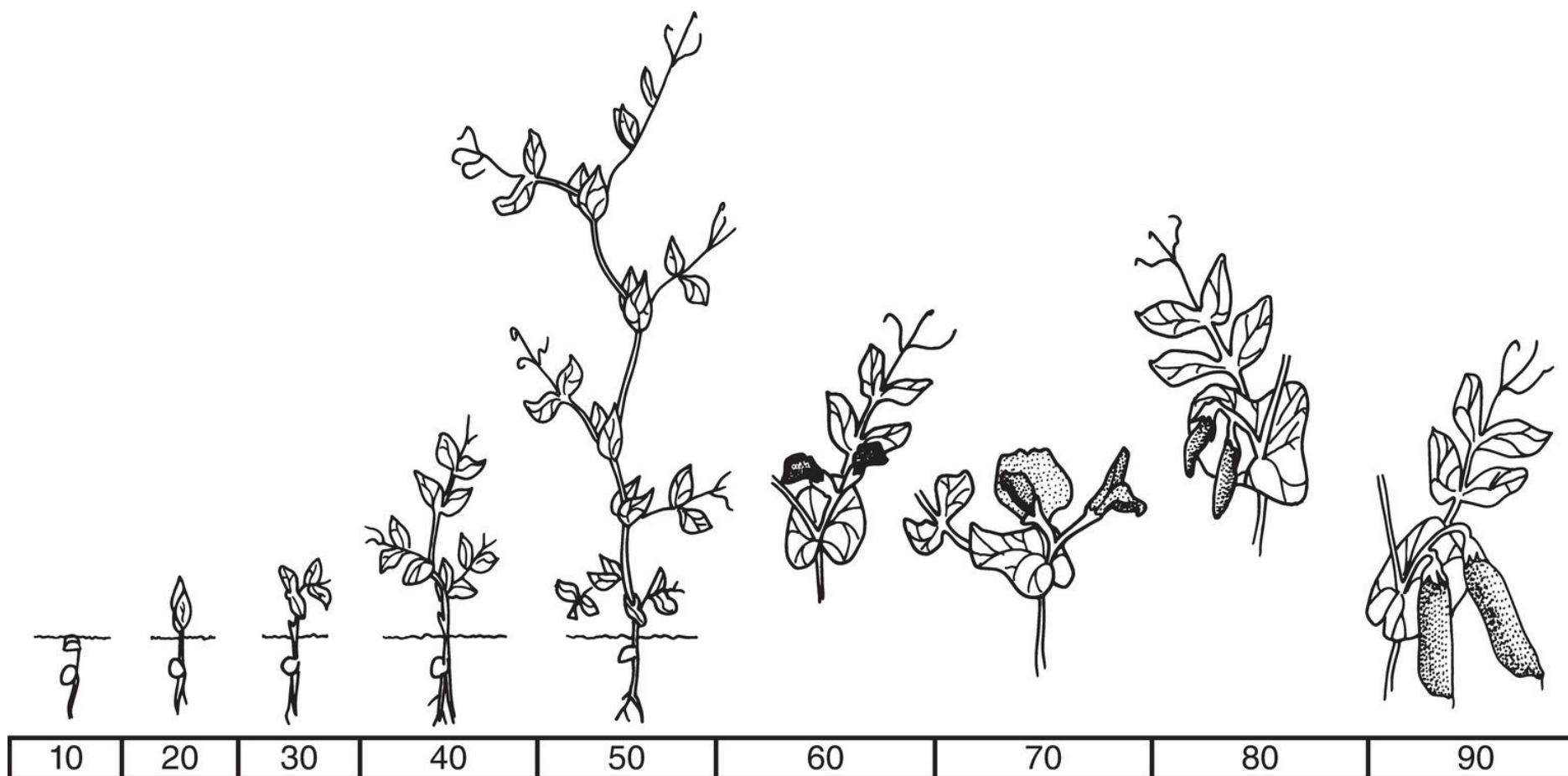


Fig. 20.9 Key elements in disease forecasting. (© L. Munk.)

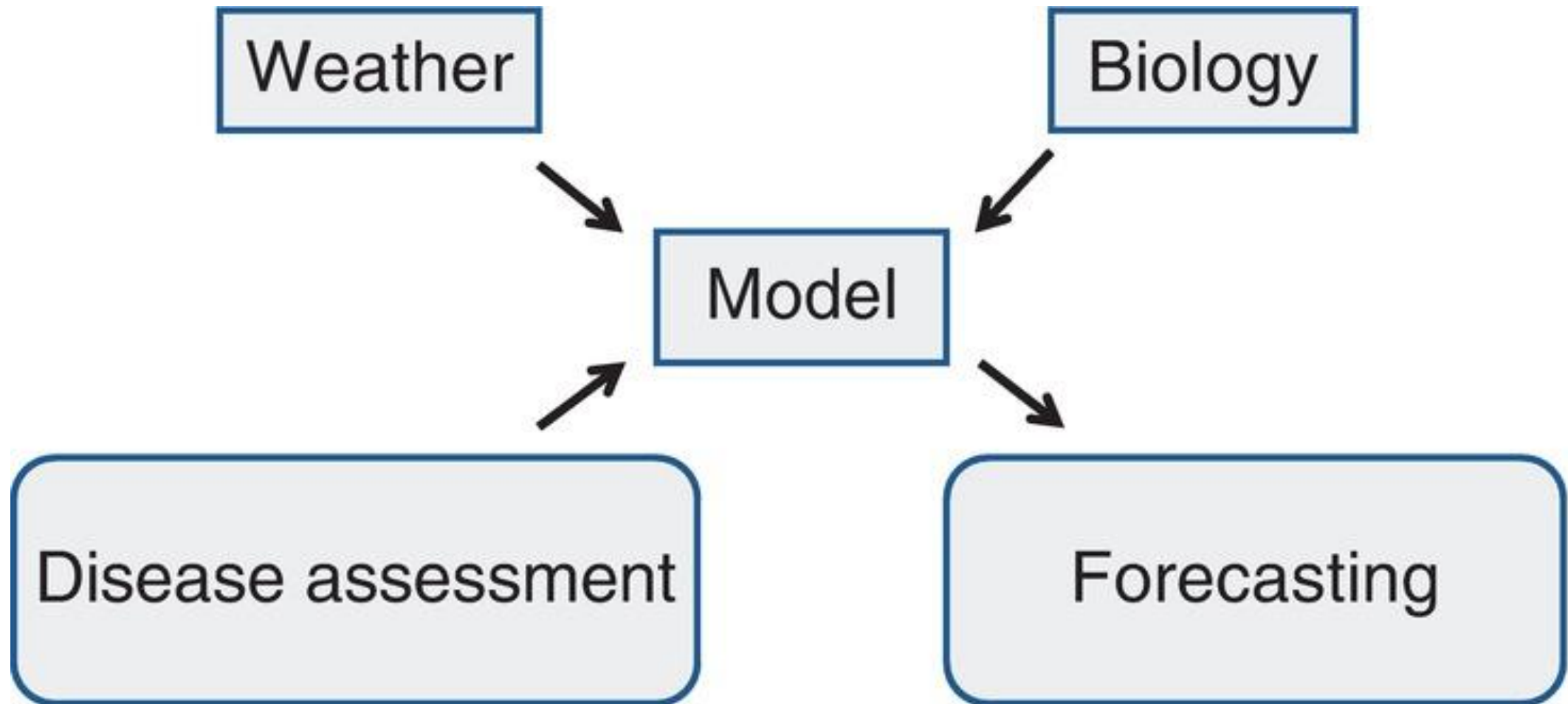


Fig. 20.10 A receiver operating characteristic (ROC) curve. (© J. Yuen.)

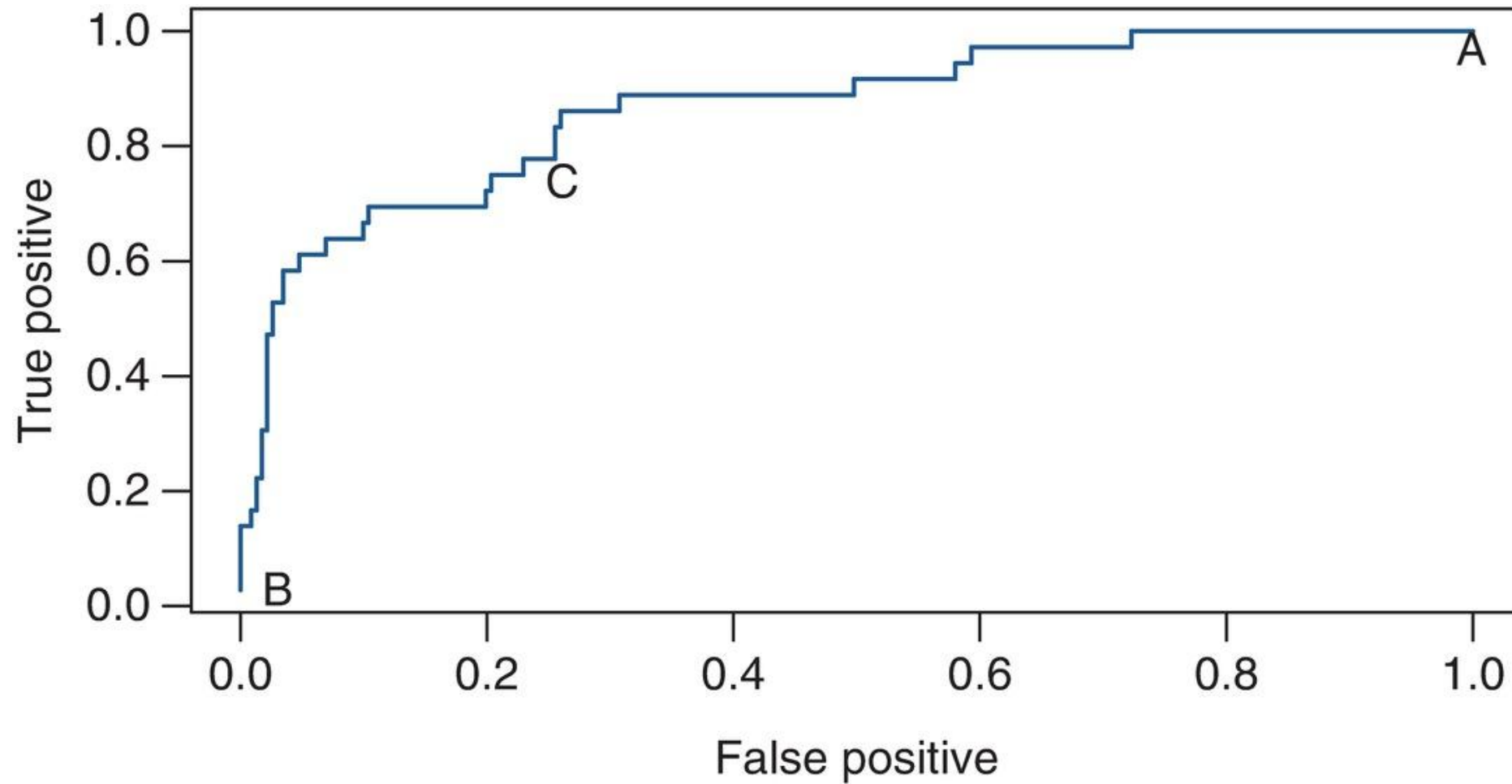


Fig. 20.11 A Burkhard spore trap collecting spores of an airborne pathogen. (© L. Munk.)



Fig. 20.12 Elements in decision support systems leading to grower's action. (© L. Munk.)

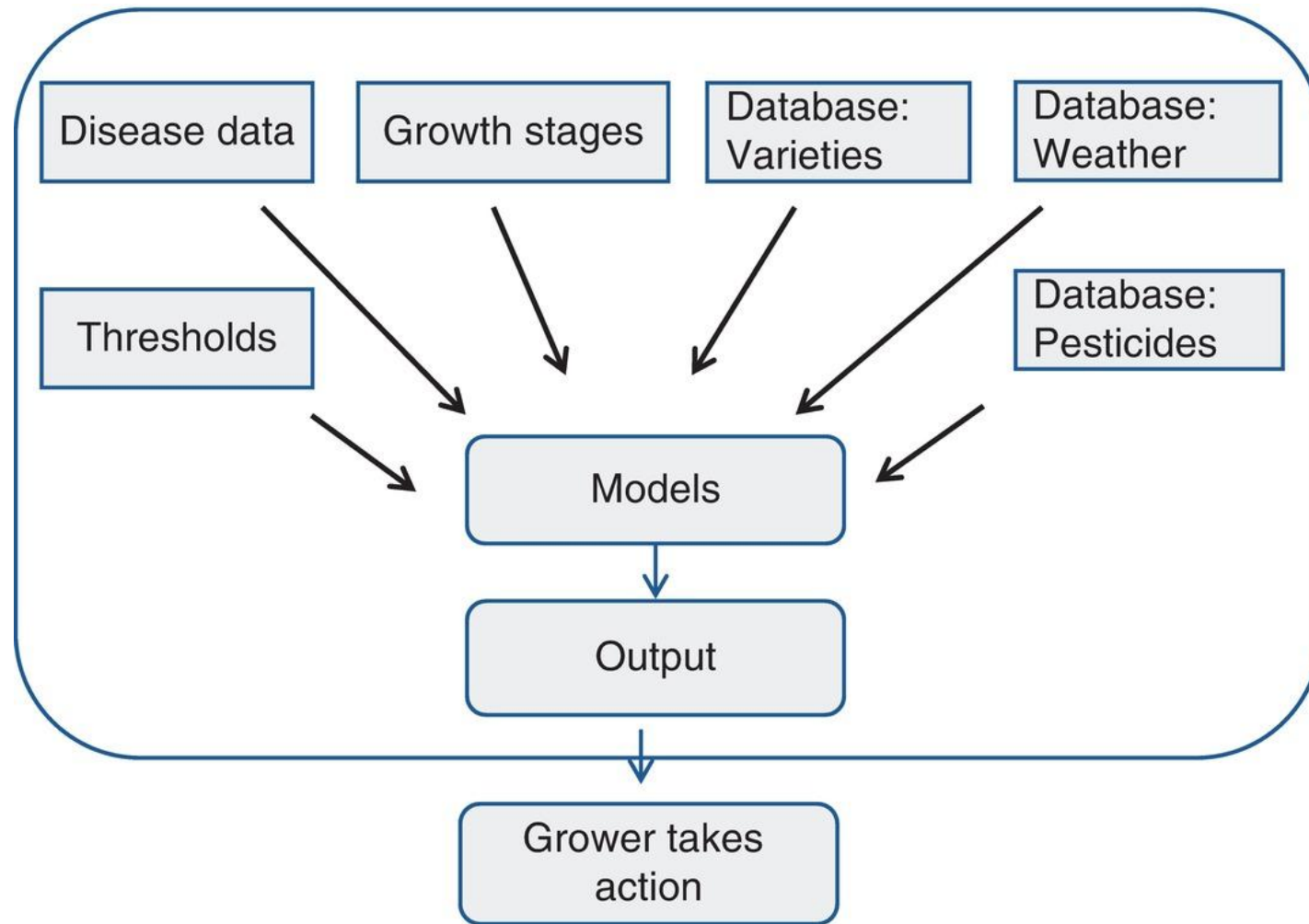
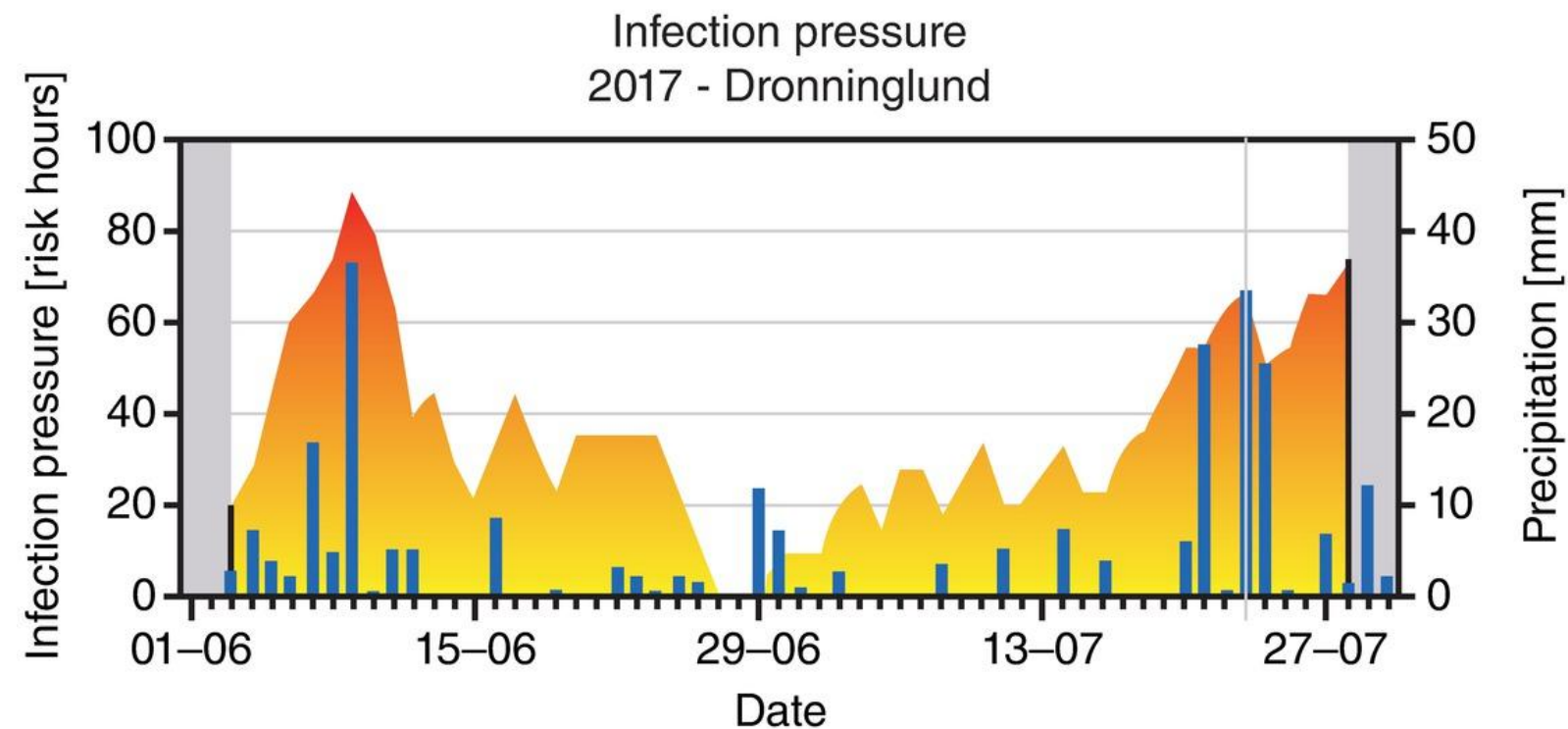


Fig. 20.13A Blight Management – Infection pressure data for Dronninglund, 2017. (Courtesy of J. Grønbech Hansen and P. Lassen.)

(A)



Recommended dose [%]	
Denmark	Region
75	100
75	100
50	100
50	75
50	50

Fig. 20.13B Blight Management – Daily risk value for Dronninglund, 2017. (Courtesy of J. Grønbech Hansen and P. Lassen).

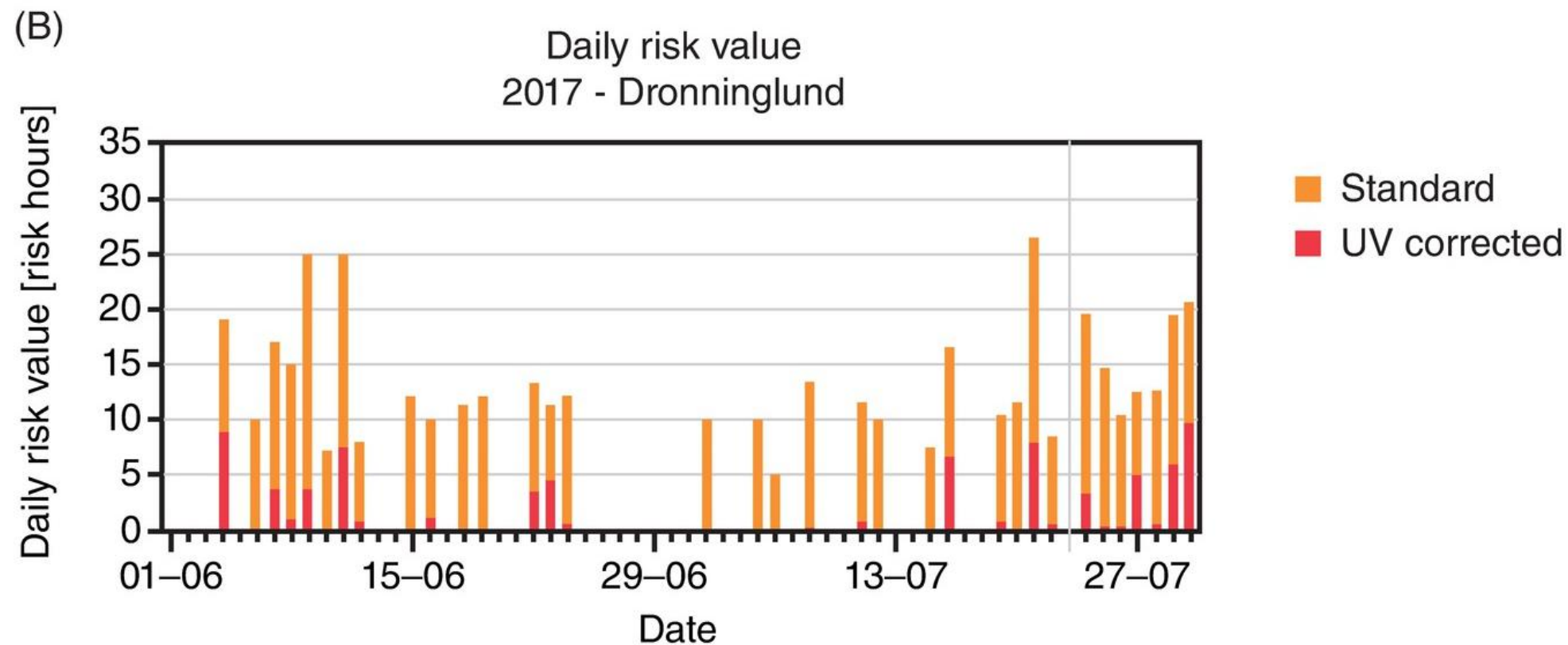


Fig. 20.13C Blight Management – Weather data for Dronninglund, 2017. (Courtesy of J. Grønbech Hansen and P. Lassen.)

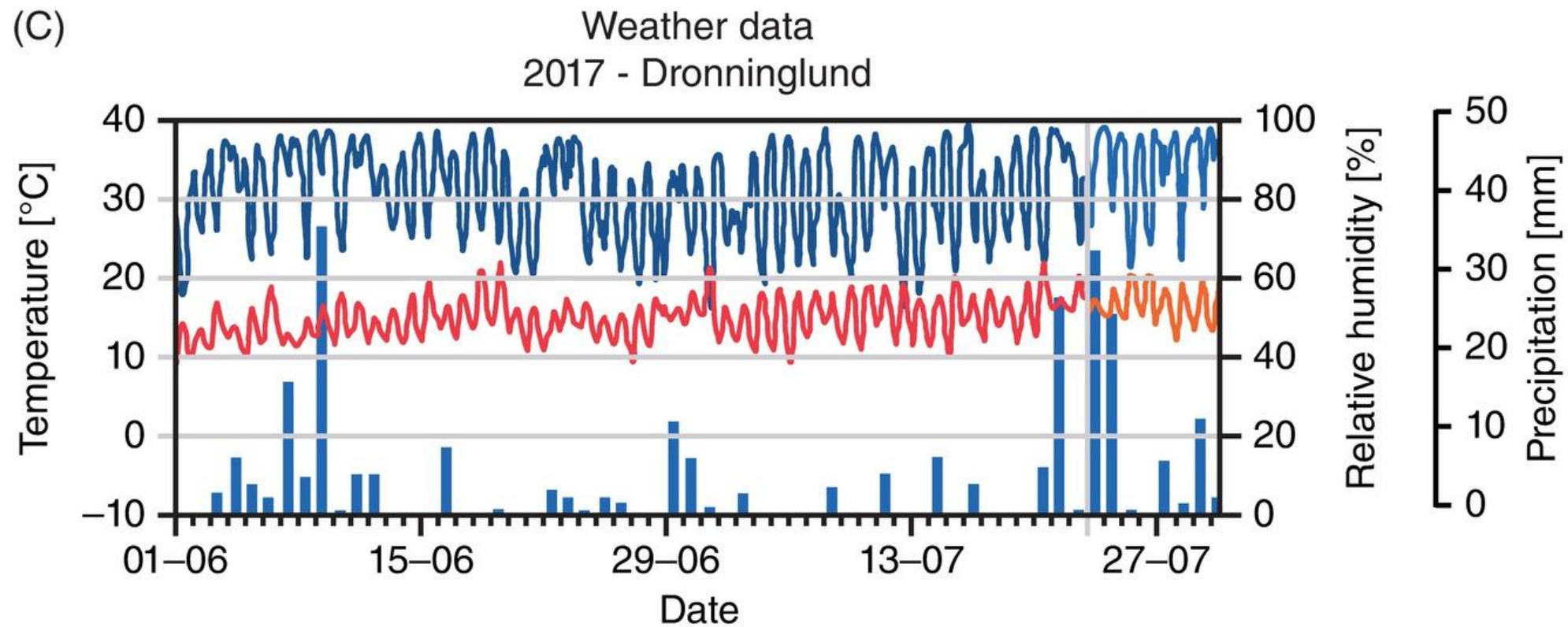


Fig. 20.14A (A) A typical example of RIMpro simulations. (Courtesy of M Trapman.)

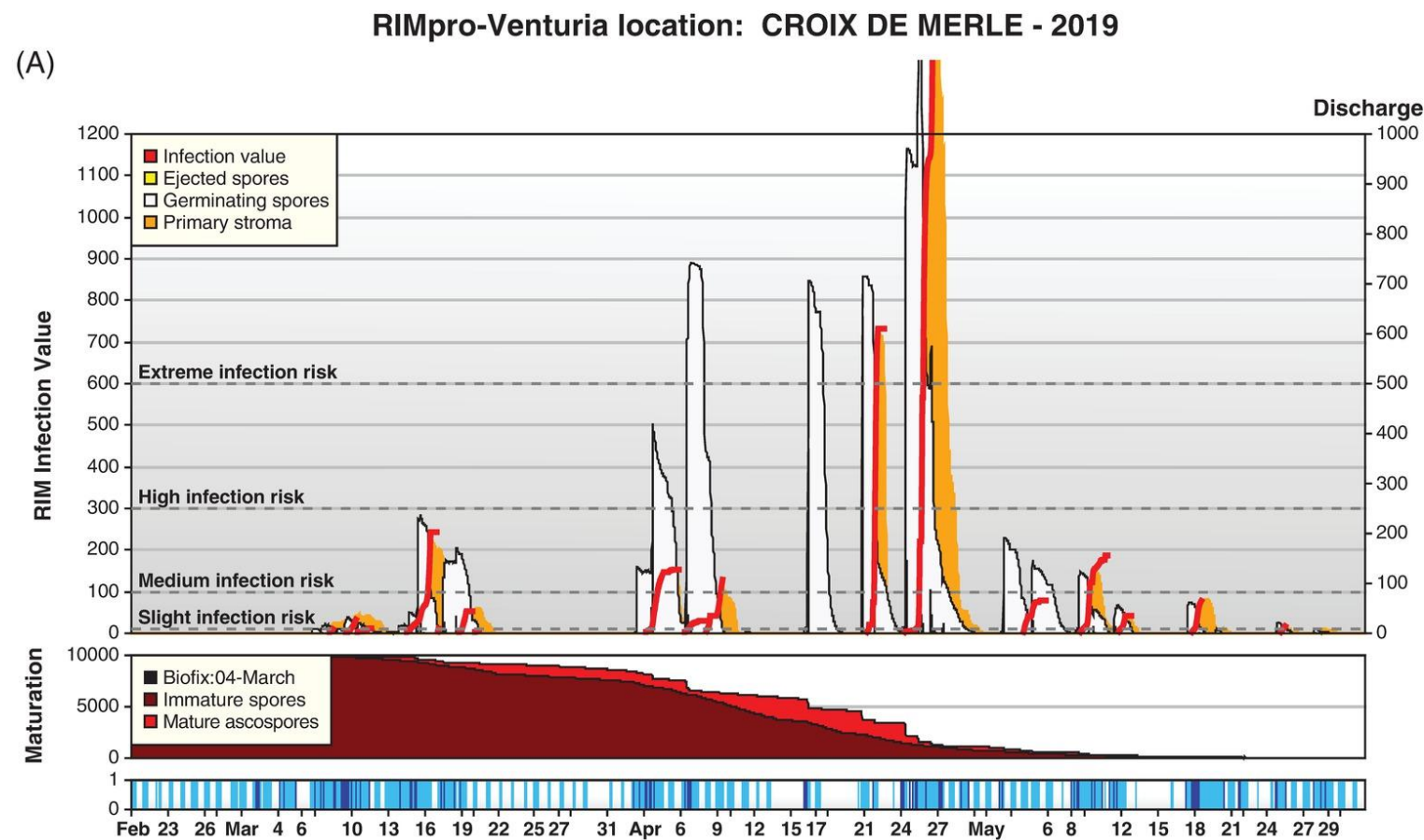


Fig. 20.14B (B) RIMpro output after fungicide application. (Courtesy of M. Trapman.)

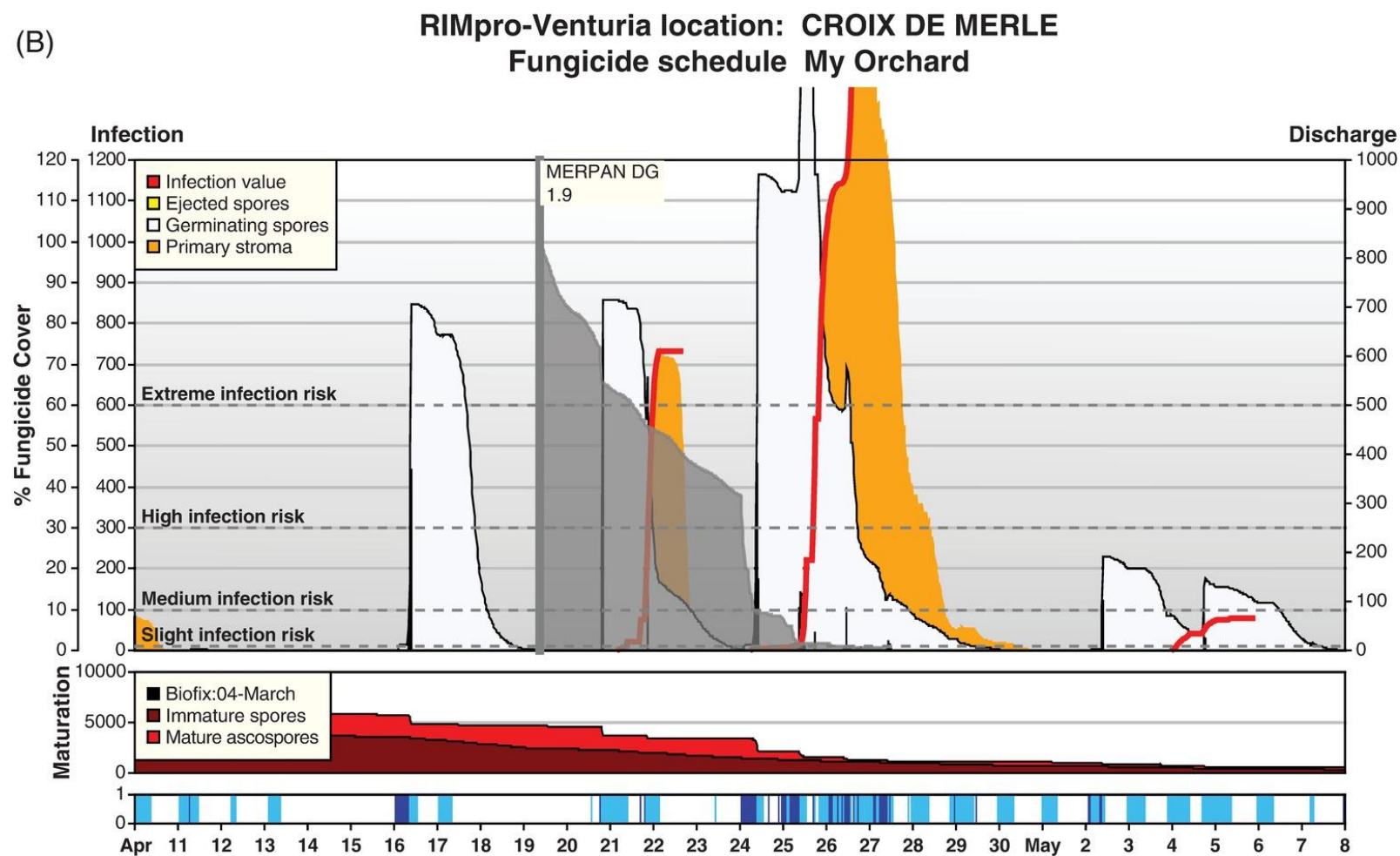


Fig. 20.15 Graphic presentation of disease progress over time with different thresholds. (....) Warning threshold; (----), action threshold; (-.-.-), damage threshold. (© J. Yuen.)

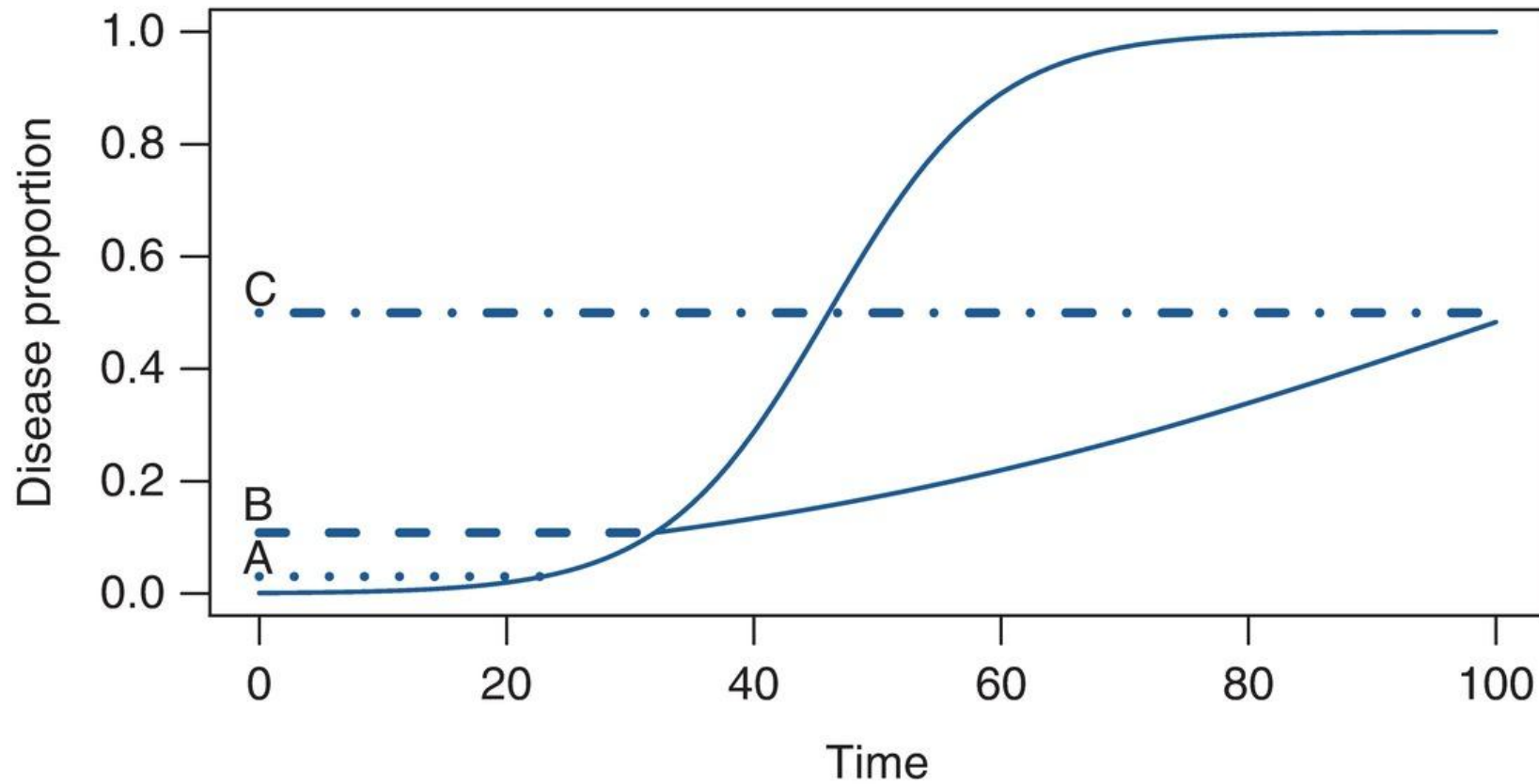


Fig. 20.16 Yield levels and yield loss. (© B. Andersson.)

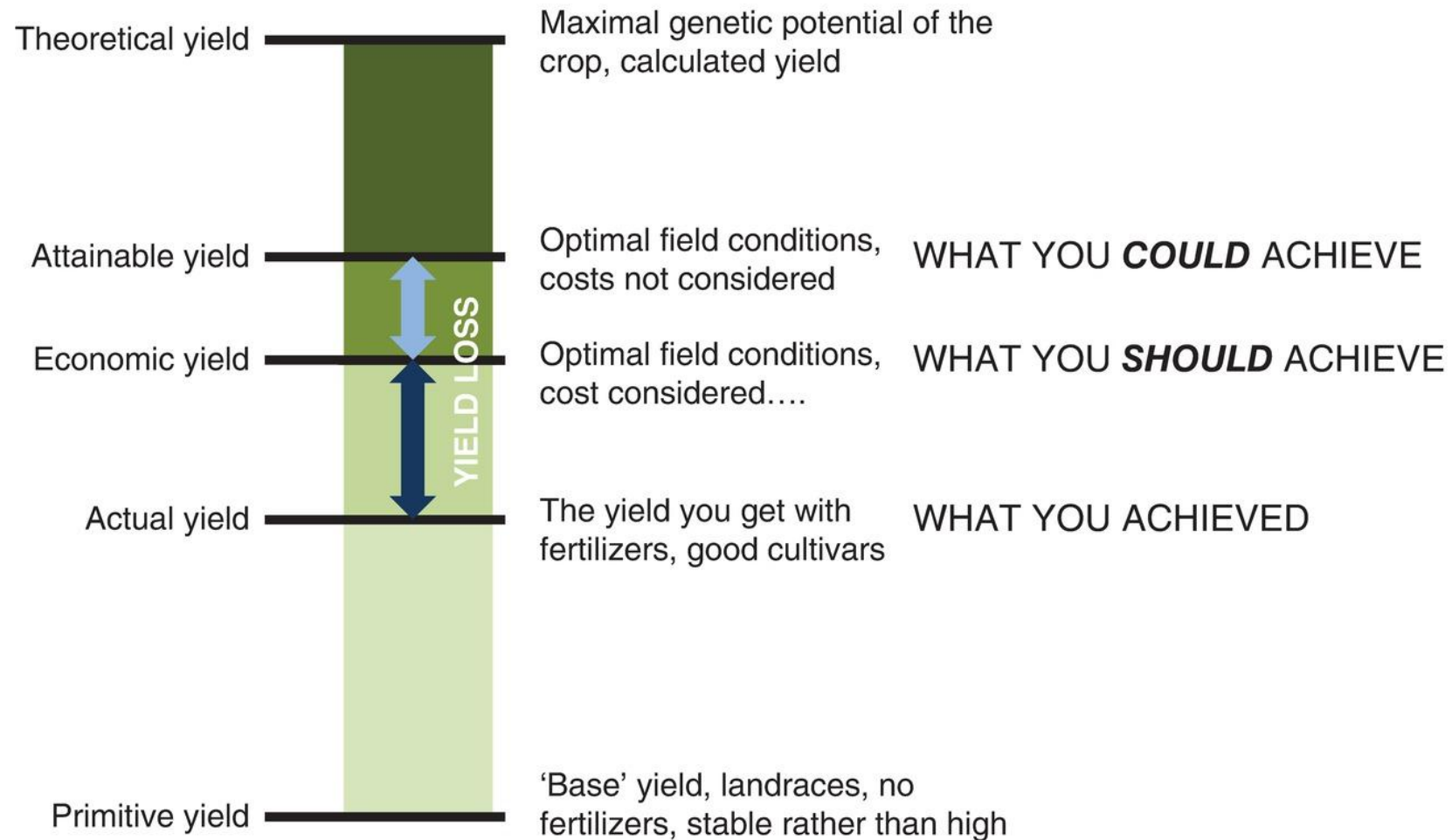


Fig. 20.17 Potatoes with and without tuber rot after sorting in storage. (© B. Andersson.)



Fig. 20.18 (A) A single point model. The dotted lines show alternatives for disease progress that would give the same loss estimate. **(B) A multiple point model.** The loss equations in the text are obtained based on the solid lines after linear transformation. (© B. Andersson.)

